

PHYSICAL AI STRATEGY

PHYSICAL AI: WHY NOW?

Multi-modal enhancements to large language models recently endowed embodied AI agents with the ability to comprehend the physical world, understand instructions via natural language, and make predictions about how actions yield changes in their environment. This new control paradigm will produce robots with near human-like capabilities for interpreting their surroundings, moving through places and interacting with objects.

While predictions about the full societal impact of embodied AI remain uncertain, one consequence is clear: businesses that rely on physical operations will face new pressures and new opportunities as humanoid robots approach commercial readiness. These machines promise a step-change in economic efficiency for tasks traditionally performed by humans, particularly in warehouse environments. For Chewy, whose ability to fulfill customer orders is core to its business, any disruption to fulfillment center (FC) operations is existential. As competitors begin to explore humanoid deployments, standing still could mean falling behind.

Ignoring this technology now risks not only missing efficiency gains but also being deprioritized by the winners of an emerging robotics market that is already forming around strategic alliances. Humanoid robotics firms are partnering with automotive manufacturers that will consume robot production. Figure.ai is working with BMW, Boston Dynamics with Hyundai, Tesla is developing Optimus for its own use. These alliances leave limited capacity for future customers. Early engagement with the market ensures Chewy's interests remains a priority when these technologies become commercially deployable.

There are many tasks in Chewy's FCs that are economically attractive and technically feasible, but picking, packing and truck loading stand out as leading candidates for humanoid robots. Each task is physically demanding or ergonomically repetitive in ways that humanoids are uniquely suited to handle; for example, lifting and maneuvering inventory during picking, executing precise movements during packing, or handling bulk weight and demonstrating spatial judgment while truck loading. And the economics are compelling: in our FCs, 2.5 million labor hours per year are spent picking, packing and truck loading; an opportunity to save \$69 million annually of variable cost by automating just these three tasks.

Moreover, the introduction of humanoids offers more than just an opportunity for incremental cost savings. It establishes the essential foundation for a future in which Chewy could operate fully lights-out fulfillment centers, with no human labor onsite. In such facilities, humanoid robots would provide the connective tissue across workflows, filling in critical gaps in material movement and task execution that cannot be fully automated by traditional robotics or conveyor systems. By integrating humanoids into today's operations, we position ourselves to bridge the last mile of warehouse automation and progress toward a future where fulfillment operations can be run entirely by machines.

WHAT ARE WE AIMING TO ACCOMPLISH?

Influence development roadmaps across the humanoid robotics industry to solve manipulation problems core to Chewy's operations and prepare Chewy for the integration of a fleet of humanoids into the FC environment.

The recent AI explosion is fueled by mind-boggling amounts of data. GPT-4 was reportedly trained on 13 trillion tokens, an internet-scale corpus of text. Stable Diffusion was trained on 5.8 billion images and Google's Lumiere was rumored to have been trained on 30 million videos. Yet there are only a handful of datasets that capture robot data in the format needed for training physical AI models. According to NVIDIA's 2025 paper, the GROOT N1 physical AI model is trained on only 3,000 hours of robots doing tasks from just 10 sources. **We believe there is an enormous opportunity to influence the future of physical AI capabilities by publishing and making freely available datasets of robots doing the specific work that Chewy cares about**—namely the manipulation of heavy deformable bags.

To be clear, humanoid handling of heavy objects remains an unsolved research-grade problem, especially when contending with deformable items that exhibit a shifting center of mass. But solving that task is so lucrative (think of moving bags of mulch, sand, or concrete mix and the myriad companies who pay for that), and there is so much investment activity in the humanoid space, that we believe humanoid robotics companies will not need much convincing to focus on this problem, especially if we lower their activation energy by providing them with tailor-made datasets.

Importantly, the success of our strategy does not require us to compete with other buyers of robots for seller’s attention, nor does it require us to partner with a robot vendor prematurely. We defer our selection until the technology matures, and we identify a vendor with an aligned go-to-market approach and compatible partnership terms.

In the meantime, there are production systems we will build to accommodate the coming fleet of humanoids in Chewy’s operations. Warehouse management systems (WMS) are responsible for assigning work to people and automation equipment, but they are fragile and not easily adapted to new classes of mechanized workers. Further, humanoid robotics companies are focused on single-agent control and treat fleet-wide decision systems as a customer concern if considered at all. **We believe there is an unmet need for an orchestration layer between the WMS and the individual humanoid robot controller that Chewy is uniquely positioned to build.** This Robot Management System (RMS) layer will manage work assigned to the humanoid fleet, route humanoids through Chewy’s FCs avoiding obstacles, and manage collective concerns of the fleet like battery charging schedules and predictive maintenance policies. The RMS will take advantage of the unique properties of a hive workforce, leveraging instantaneous communication to propagate newly discovered grasp strategies or observed changes to the environment from one humanoid to its siblings across the network.

Whereas our strategies may evolve as we learn, we hold the following tenets as our durable statements of purpose:

1. **Train the World’s Robots on Chewy Work.**

We aim to influence physical AI and embodiment roadmaps across the robotics industry to consider the class of manipulation tasks common in Chewy’s FCs. We will build and operate a laboratory to produce data of robots doing routine FC tasks such that all humanoid companies will find our data indispensable for training their AI models. In doing so, prospective robots we consider deploying in the future come to us pretrained on our data.

2. **Adopt When Ready, Not When Flashy.**

Chewy will adopt proven commercial humanoids whenever they meet our requirements and when economics align. Robots will enter production only when their fully loaded cost performs better than human benchmark and demonstrates a clear path to future savings over time. This standard will tie adoption to measurable business outcomes rather than novelty.

3. **Lead with Technical Credibility in Physical AI.**

We aim to become recognized experts in simulation, control and experimental validation of humanoid robots and mobile manipulators with emphasis on generative AI for physical embodiments, especially reinforcement learning of multi-modal models. We target conference and journal publications, public talks, social media posts and featured partner status at industry events; and we should fund academic engagements that further research in deformable solid modeling techniques. We should default to being open, aiming to publish datasets, digital twin models, and performance metrics.

4. **Partner Broadly, Stay Top of Mind.**

We must establish non-exclusive development partnerships with both emerging and mature companies working in the space and foster relationships so that we are top of mind when they set their roadmaps and finalize design decisions.

5. **Build for Fleet-Scale Robotics from Day One.**

We must lay groundwork today so that Chewy’s warehouse operations are primed to incorporate a fleet of interconnected robots when timing dictates. We aim for robots that work collectively, federate learnings among themselves, share knowledge of their environment with each other and seamlessly balance workloads across the fleet. We focus only on the components of an orchestration stack that Chewy Robotics can uniquely provide and ignore the undifferentiated heavy lifting of a general warehouse management system (WMS) and the vendor-proprietary single-agent control software.

HOW DO WE MEASURE PROGRESS?

Key Result 1: Publish a high-quality, open-source *dataset* of humans and humanoids manipulating deformable, heavy bags and the corresponding *simulation artifacts*.

The dataset must be undeniably useful for any company training physical AI foundation models such that its inclusion of our data in their training sets is a no-brainer. We must meet the robotics companies where they are by producing curated data with attributes they expect, in formats that are compatible with their development tools, hosted on platforms they use for data consumption, and documented with domain expertise that demonstrates our credibility. Any robot trained on this dataset then arrives to us with innate abilities to do the tasks we care about, commoditizing the capability and providing flexibility in our future partner selection.

Publishing high-quality training data requires us to become great in three essential areas.

Reinforcement Learning: We must become great at employing reinforcement learning to fine tune physical AI foundation models, wherein we evaluate candidate datasets from our simulations and experiments for their suitability for publication. Proving to ourselves that the data we collect are essential for others to train physical AI models with the capabilities we ultimately want to purchase on the open market, ensures we recognize the standard and quality with which physical AI training data must be curated. In doing so, we will gain new perspective on the specific tasks we target and the evolving state-of-the-art.

Testing: We must become great at experimental evaluation of both humans and humanoids in common tasks within Chewy's FCs. We must duplicate FC infrastructure in laboratory conditions, add measurement instrumentation and experimental controls to the point where repeatable science can be performed, and digital twins can be calibrated. In doing so, we will understand the limits of human performance doing tasks of interest so that we can judge when robotic technology is sufficiently mature.

Simulation: We must become great at simulation and the creation of digital twins of the processes of interest. Training AI systems is data-intensive, and direct measurements from our test facilities will not be sufficient to produce data at the scale or with the diversity that a robust physical AI model requires. Digital twins capture essential physics, enable domain randomization, and simulate interactive scenarios such that causal relationships between robot action and object response can be robustly learned. Developing this competency must include advancing the state-of-the-art in deformable physics modeling and in the soft contact mechanics of end effectors—essential skills for robots in FC contexts. Further, simulation environments must represent the optical properties of objects and lighting in the FCs with sufficient fidelity to train vision models.

Key Result 2: Publish a Robot Management System (RMS) architecture¹ to orchestrate a fleet of humanoids.

Humanoid robot companies are scope-locked on developing the capabilities of an *individual* robot and only seem to invest in single-agent use cases. There is little evidence that these companies think about the aggregate capabilities of a *fleet* of humanoids, how to operate them as a collective, how to integrate them into existing infrastructure, how to federate learnings, and how to leverage zero-cost communication among the population. Further, when deployed into a fulfillment center, the fleet must be deeply integrated with Chewy's warehouse management system (WMS) and computerized maintenance management system (CMMS).

By leveraging zero-cost, instantaneous communication across the humanoid fleet, we unlock capabilities that are not possible with individual robots or human workers. Shared experiences allow for federated learning, where lessons acquired by one robot (such as optimal grip strategies) are instantly propagated to all others. Real-time task reallocation becomes feasible through decentralized coordination, reducing idle time and increasing throughput. Robots can collaboratively construct a shared perception of their environment, improving vision model robustness in occluded or

¹ An architecture does not imply a functional and validated codebase. Once an architecture defines the RMS scope, system requirements, interface descriptors, software component hierarchy and prototype algorithms, the project may be handed off to a software development contractor or developed in-house through heavy use of LLMs.

variable lighting conditions. Communication also enables distributed predictive maintenance, as robots can monitor the operational health of peers and proactively adjust workloads or flag issues. Finally, coordinated safety protocols can be enacted instantaneously, ensuring a synchronized response to hazards in mixed human/robot environments.

Publishing an orchestration architecture requires developing expertise in two critical areas.

Safety: We must become great at designing, simulating, and validating safety-critical behaviors for humanoids operating alongside humans. This includes developing real-time monitoring systems that detect and respond to hazardous conditions, ensuring failsafe execution of motor commands, and integrating physical redundancies (e.g., compliant actuation, soft padding) with intelligent control strategies. By instrumenting our lab environments with sensing infrastructure such as LiDAR, stereo cameras, and wearable trackers, we can analyze near-miss incidents, simulate edge-case failures, and systematically validate that our safety protocols exceed OSHA and ANSI standards. In doing so, we earn trust from our team members and readiness for future regulatory environments.

Algorithms: We must become great at developing algorithms for autonomous coordination of a heterogeneous robot fleet. This includes multi-agent task scheduling under dynamic workloads, real-time path planning in semi-structured environments, and optimizing energy usage across long shifts. We also need telemetry-based models to predict component wear, trigger preventative interventions, and enable federated learning to propagate improvements across the fleet. In doing so, we move beyond basic robotic automation to a high-efficiency system that dynamically adapts to Chewy’s operational cadence, minimizes downtime, and delivers compounding returns as the fleet scales.

HOW DO WE COPE WITH MASSIVE UNCERTAINTY?

Dexterity, strength, and mobility are the three fundamental physical attributes required for humanoid robot performance—and they exist in tension. No single robot, in the foreseeable future, will achieve human-level capability in all three dimensions simultaneously. Instead, each robotics vendor makes a deliberate tradeoff in this design space, optimizing for one or two attributes at the expense of others. Fortunately, Chewy’s fulfillment centers offer a range of economically attractive tasks that span different combinations of these capabilities. For example, packing tasks demand high dexterity but little strength or mobility; truck loading requires significant strength but minimal dexterity; and picking involves high mobility and strength with less need for fine manipulation.

	Pack Station	Truck Loading	Pick Mod
Dexterity	High	Low	Low
Strength	Low	High	High
Mobility	Low	Low	High

By pursuing multiple tasks across this spectrum, we create a strategic hedge against uncertainty in the pace and direction of robotics development. Through our partnerships, we gain early insight into each vendor’s strengths and can steer them toward the use case where their embodiment choices are most likely to deliver value. Should the field evolve in an unexpected direction—favoring, for example, mobile over dexterous systems—we retain the flexibility to pivot toward tasks with compatible requirements. This modularity of opportunity is a strength, not a limitation.

Further, we remain open to pragmatic workflow changes that preserve economic value, such as adding an integrated barcode scanner to a robot’s end effector or changing rules in the WMS to direct easier picks to humanoids. This approach avoids locking ourselves into irreversible infrastructure changes. In this way, we maintain both strategic optionality and operational discipline as the technology matures.

2025 ANNUAL PLAN

Hiring Summary

The Robotics Team currently has an allocated headcount of 11 with seven positions filled, leaving four available to backfill. In 2025, we plan to grow the allocated headcount by two and fill the four budget-neutral backfills.

	Current	2025
Allocated	11	13 (+2)
In-seat	7	13 (+6)

The skills of the six new hires will target reinforcement learning, multi-agent algorithms, simulation and experimental test.

Level	Title	Skills	Notes
C08	Principal Research Scientist, Physical AI	Reinforcement learning/imitation learning to fine-tune foundation models	New
C08	Principal Research Scientist, Algorithms	Multi-agent planning, SLAM, control architectures, estimation theory	New
C07	Associate Director, Simulation	Digital twin, Omniverse, small team leadership	Backfill
C06	Senior Test Engineer	Instrumentation, lab ops, experiments owner	Backfill
C06	Senior Robotics Scientist, Physical AI	Support RL & SFT dev	Backfill
C06	Senior Robotics Scientist, Algorithms	Optimization, multi-agent support	Backfill

Budget Summary

Operating Expenses		Expected Timing
AWS (GPU compute + storage)	\$200k/yr	May 2025
Robot Lease (Apptronik, Reflex, etc)	\$100k/yr	Aug 2025
Data labeling consulting	\$50k/yr	Aug 2025
Software Licenses (Omniverse, MATLAB, etc)	\$25k/yr	Jun 2025
Workforce		
New hires	+2 HC	
Travel, conferences, etc.	\$40k/yr	
OPEX Total	\$415k/yr, +2 HC	
Capital Expenditures		
Robots		
UniTree G1, H1-2, Go2 (3x) robots	\$230k	May 2025
Chewy BUSTER humanoid torso robot	\$220k	Aug 2025
TBD Future humanoid robot purchase	\$150k	Nov 2025
Replacement Parts	\$50k	Dec 2025
Instrumentation & Lab		
Motion capture and object tracking system (OptiTrack)	\$100k	Oct 2025
Teleop HW, XR goggles, LiDAR, wearable trackers, etc.	\$100k	Oct 2025
General lab tools, equipment, facility improvements	\$100k	Nov 2025
Laptops with NVIDIA GPUs	\$60k	Jul 2025
CAPEX Total	\$1,010k	
TOTAL ANNUAL BUDGET		\$1,425k, + 3 HC

PROJECTS

Project 1: Data Publication Effort

Project Objective: Publish a dataset consisting of (1) 200 hours of humans and humanoids completing tasks in the physical TREAT Lab, (2) 20,000 hours of domain-randomized examples in the simulated FC environment, and (3) associated simulation models and artifacts of the environment, robots, and manipulated objects. Humanoid robots are to be acquired through purchase agreements (e.g., UniTree H1-2), leased from partners (e.g., Reflex Robotics), and built in-house (e.g., Chewy's BUSTER humanoid torso). (See Risk 2)

Constructing physics-based simulations of *rigid bodies* at sufficient fidelity for RL is becoming common practice and is (or will be) within our capabilities. However, building such models of *deformable* objects is still an open research problem, but is receiving increased attention from academics, robotics companies, and tool vendors. (See Risk 3)

Key Results:

1. Validated digital twin models of the PickMod with rigid objects and at least one humanoid.
2. 10-hour simulation with domain randomization.
3. Validated digital twin models of humanoid robots with simulated control logic.
4. 100-1,000 hours of simulated data used for RL training.
5. 10,000-20,000 hours of simulated data used for RL training.
6. Validated high fidelity model of deformable bags.
7. Publish dataset consisting of 200 hours of physical, 20,000 hours of simulated tasks and associated simulation models and artifacts. PR blitz.

Project Tasks:

1. Simulation Models & Digital Twins

- Build and validate digital twins of three FC operations (PickMod, pack station, truck load).
- Develop high-fidelity simulation models of deformable objects, including heavy bags.
- Create photorealistic and physically accurate models of robots and objects for use in Omniverse.
- Integrate domain randomization techniques into simulated environments for robustness.
- Align simulation outputs with instrumentation data from TREAT Lab to support calibration and benchmarking.
- Publish simulation artifacts and documentation to support reproducibility and adoption by external partners.

2. Robot Control & Physical AI

- Develop and test BUSTER humanoid torso robot, including hardware integration and low-level control software.
- Build modular control stack (teleop, scripted, autonomous) for BUSTER, reusable across partner platforms.
- Fine-tune open-weight foundation models using reinforcement learning in Chewy's simulated environments.
- Evaluate performance of RL policies in simulation and transfer to physical platforms.
- Integrate motion planning, perception, and actuation components for humanoid manipulation tasks.
- Implement data logging and control instrumentation interfaces for all robot platforms.

3. Experimental Work

- Define standardized experimental protocols and performance metrics for human and robot trials.
- Conduct baseline performance evaluation with human operators on deformable object tasks.
- Execute teleop and autonomous trials using BUSTER and partner humanoids in the TREAT Lab.
- Collaborate with partner robotics vendors to integrate and test external humanoid platforms in TREAT Lab.
- Label and organize collected data to prepare for publication, including metadata on task, agent, and environment.

Project 2: TREAT Lab (Testing & Research of Embodied AI Technologies)

Objective: Construct a comprehensive, instrumented laboratory replicating key FC stations to support rigorous experimentation and scalable data collection.

The purpose of the TREAT Lab is to provide a controlled, instrumented environment where repeatable experiments can be conducted to evaluate both human and robotic performance on fulfillment center tasks. It consists of modular replicas of key FC workflows, including PickMod, pack stations, and truck loading zones, equipped with advanced sensing and data capture systems. It supports high-fidelity instrumentation, flexible task setups, and standardized experimental protocols,

ensuring consistency across trials. This environment is essential for validating digital twins, calibrating simulation models, and generating training data for reinforcement learning.

We estimate data collection in the PickMod from *human* subjects to be routine and low risk, from *teleop'd humanoids* to be achievable with standard engineering practices, and from *autonomous humanoids* as at the frontier of robot control and only achievable by the most sophisticated practitioners (See Risk 1).

Key Results:

- 1. TREAT Lab Commissioning & Ribbon Cutting.
- 2. 6hr test of **human** in PickMod.
- 3. 6hr test of **teleoperated humanoid** in PickMod.
- 4. 6hr test of **autonomous humanoid** in PickMod.

Project Tasks:

- 1. PickMod Installation and Experimental Infrastructure
 - Partner with FC General Contractor to design, install, and commission PickMod.
 - Construct experiment fixture systems for continuous material flow, including replenishment, and output collection.
- 2. Instrumentation & Sensing Systems
 - Procure and install sensing hardware: OptiTrack motion capture, LiDAR, XR interfaces, and force sensors.
 - Run pilot studies with human participants to validate instrumentation fidelity.
- 3. Experimental Design & Protocols
 - Define and implement repeatable test procedures for human, teleoperated, and autonomous operation.
 - Develop baseline scripts and task scenarios for human and robot trials.
 - Design and construct systems for safe teleoperation of humanoid robots, including a gantry or harness system to support balance and fall risk mitigation.
 - Configure safety systems for robot operation, including soft barriers and emergency stop mechanisms.
 - Train team members on lab usage, safety, and experimental methodology.
- 4. Data Pipeline Development
 - Build cloud-based infrastructure for real-time ingestion, structured labeling, and scalable data storage.

Project 3: Robot Management System (RMS) Algorithms

Objective: Design and prototype an orchestration system for a humanoid fleet operating in dynamic warehouse environments. This includes task allocation, real-time routing, safety systems, and federated learning—all integrated with WMS/CMMS systems.

This project defines the architecture and develops baseline implementations of critical algorithms needed to coordinate a heterogeneous fleet of humanoids working alongside humans and receiving instructions from Chewy’s WMS. This effort emphasizes fleet-level intelligence over single-agent autonomy, and anticipates unique challenges such as shared mapping, learning across robots, and safety in mixed human/robot environments. The RMS must enable the robots to work as a cohesive unit, dynamically adapting to workload demands, environmental changes, and system health conditions. The sequence of algorithm development will be determined as an outcome of the architecture review, and include:

Task Allocation & Scheduling	Algorithms for dynamic assignment of tasks across a robot fleet based on priorities, proximity, capabilities, and workload balancing.	TRL ² 6
Fleet-Level Safety & Monitoring	Real-time sensing, hazard detection, and response systems to ensure safe operation among humans and other robots.	TRL 5
Multi-Agent Path Planning	Routing strategies for humanoids in semi-structured, dynamic environments while avoiding collisions and congestion.	TRL 6
Shared Mapping & State Estimation	Development of shared world models to enable distributed decision-making and perception robustness.	TRL 3

² Technology Readiness Level. TRL 1-3 immature/requires research, TRL 4-6 some elements proven, TRL 7-9 near ready for production

Federated Learning Infrastructure	Systems to propagate learned improvements across the fleet from individual robot experience.	TRL 2
WMS/CMMS Integration	Interfaces and data pipelines that connect the RMS with warehouse operations and maintenance workflows.	TRL 9

Key Results:

- RMS architecture documented and reviewed.
- RMS first algorithm prototype demonstrated in simulation.
- RMS first algorithm prototype submitted for publication (to e.g., ICRA, NeurIPS)

Project Tasks

- Define modular RMS architecture with clearly scoped subsystems and interfaces to WMS and CMMS.
- Implement baseline task assignment algorithms and evaluate in simulation.
- Prototype shared mapping framework using lightweight SLAM techniques.
- Design multi-agent path planner that handles dynamic obstacles and priorities.
- Simulate safety edge cases and validate system-level responses.
- Implement fleet-level federated learning prototype using simulated data.
- Build integration layer with WMS for real-time task ingestion and status updates.
- Test coordination logic in simulated fulfillment center environment.
- Document API contracts between RMS subsystems and external systems.

MILESTONES

	QBR	Milestones Reported at QBR	Capability Unlocked
Yr 1	June '25	Physical AI Strategy & Roadmap	Establishes shared vision and strategic rationale for Physical AI efforts. Aligns team and leadership around priorities, partners, and investment areas.
	Sept '25	TREAT Lab Commissioning & Ribbon Cutting	Unlocks the experimental environment for physical data collection.
		Validated digital twin models of PickMod with rigid objects and one humanoid	Provides simulation environment for scalable reinforcement learning.
		10-hour simulation with domain randomization	Domain randomization unlocks sim-based RL.
	Dec '25	6hr test of human in PickMod	Validates lab configuration, fixture systems, instrumentation, data collection pipelines. Unlocks evaluation of robotic technology.
		RMS architecture documented and reviewed	Provides clarity on algorithm functional scope, problem definitions, intended use cases. Identifies first alg. to focus on.
		Validated digital twin models of humanoid robots with simulated control logic	Provides robot platform and realistic decision behavior for RL.
		100-1,000 hours of simulated data used for RL training	Demonstrates simulation artifacts achieve meaningful scale.
	Mar '26	6hr test teleoperated humanoid in PickMod	Validate robot reliability and embodiment suitability for task. Long teleop experiments unlock path to collecting 200 hours of data for publication.
		Fine-tune an open weight physical AI model in Chewy's simulated environment	Demonstrates simulation artifacts are fit for purpose.

RMS first algorithm prototype demonstrated in simulation, submitted for publication

Yr 2	6hr test autonomous humanoid in PickMod	Requires robust physical AI model fine-tuned on Chewy's tasks. Proves viability of autonomous humanoids for FC work.
	10,000-20,000 hours of simulated data used for RL training	Demonstrates simulation artifacts are ready for publication.
	High fidelity model of deformable bags	Unlocks scalable grasp planning research.
	Publish dataset of 200 hours of physical, 20,000 hours of simulated tasks and associated simulation artifacts.	Produces the artifact that Chewy advertises to the Physical AI industry via papers, talks, etc.
	PR blitz advertising Chewy dataset. Conf/journal pubs, social posts, etc.	Influence development roadmaps across the humanoid robotics industry.

RISKS AND MITIGATIONS

1. Immaturity of Long-Duration Autonomy in Humanoids

Risk: No humanoid robot available today has demonstrated the ability to operate autonomously over long durations in real-world, unstructured environments. It is uncertain when the field will reach the necessary maturity; progress could occur within a year or two, or it may take five.

Mitigation: Chewy Robotics will pursue autonomous humanoid operation as an ambitious stretch goal, but not as a prerequisite for success. We will fine tune open-weight physical AI models for task-level autonomy and integrate them into our own experimental platforms. However, our core strategy centers on generating high-quality datasets that enable the training of such models. These datasets remain valuable even if collected from teleoperated humanoids or human demonstrations. In this manner, we retain the ability to drive industry impact and progress toward automation without depending on full autonomy in the near term. Should we succeed in demonstrating autonomous long-duration operation, it will represent a breakthrough milestone, but the strategy is designed to create value even without it.

2. Vendor Constraints and Lock-In

Risk: Our ability to meet our data publication objective is subject to purchasing robots from companies who want to tie partnership agreements to order commitments, or who may not share our perspective on open sourcing data from their robot.

Mitigation: Build our own experimental platforms (e.g., BUSTER) as a critical hedge against single point-of-failure hardware, changing whims of partner engagement terms & conditions, and ensures access to low level robot data vendors may consider proprietary.

3. Simulation Fidelity Challenges

Risk: Inaccurate models of deformable objects prevent effective transfer of simulation-learned behavior to the physical world. Additionally, finetuning humanoid robots in simulation to an acceptable performance may require knowledge of physical properties of components which may be difficult to obtain.

Mitigation: Invest early in validating digital twins and deformable object models with high realism. Iteratively evaluate reinforcement learning models on physical robots. Build knowledge in sim2real and real2sim transfer techniques and how to characterize physical robotics systems. Partner with NVIDIA on prototyping their new integration with the renowned MuJoCo physics engine. Join the Alliance for OpenUSD, and industry working group solving gaps in physics models.

4. Talent Scarcity in Core Domains

Risk: Recruiting experts in RL, simulation, and algorithms is competitive.

Mitigation: Maintain a high hiring bar, promote the technical ambition of the team, and leverage open publication as a recruiting signal.